

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

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Where cancers are caught, and whether anyone looks

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Target application

The chip

Analog front end

Operating modes

Analog blocks

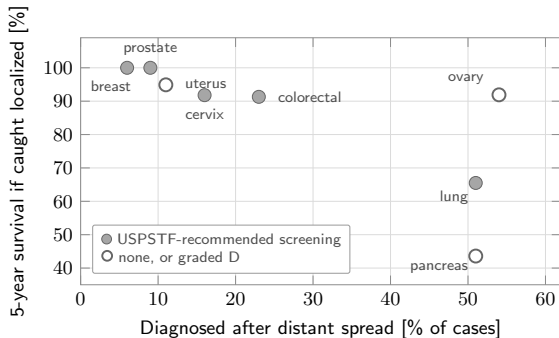
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Against Myshkin and the field

From currents to a score

References

Backup



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why ovarian: late, lethal, and unscreened

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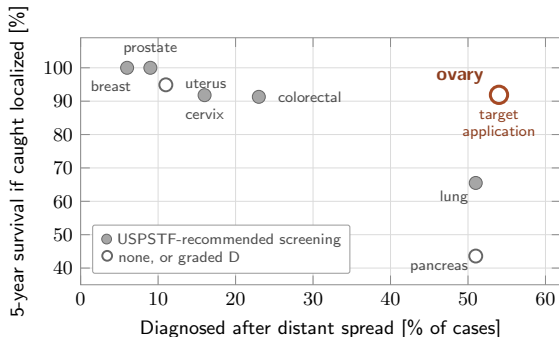
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Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

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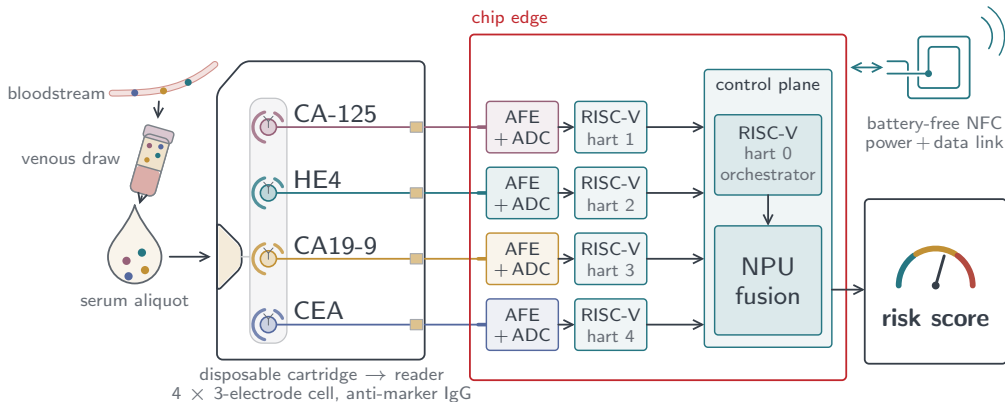
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Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single marker is a test; four fused are [5]

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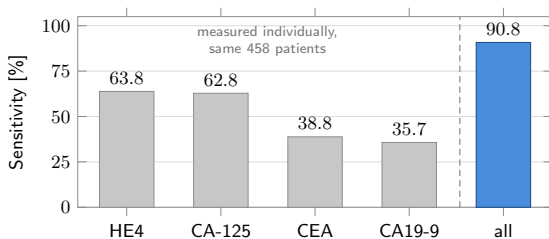
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Marker	Molecule	Cut-off
CA-125	MUC16 mucin epitope [8]	35 U/mL
HE4	WFDC2 glycoprotein [6]	70 pmol/L
CA19-9	sialyl-Lewis ^a antigen	37 U/mL
CEA	CEACAM5 glycoprotein	5 ng/mL

Each marker covers a subtype the others miss. The decision is a learned combination, so the chip is four channels plus the processor that fuses them.

Block diagram

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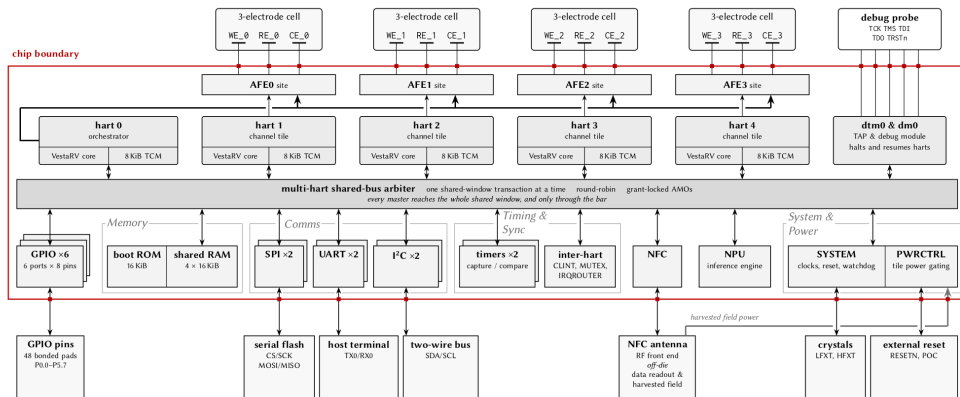
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Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

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ARTWORK PENDING



**ARTWORK
PENDING**



Grounded-WE two-amplifier potentiostat

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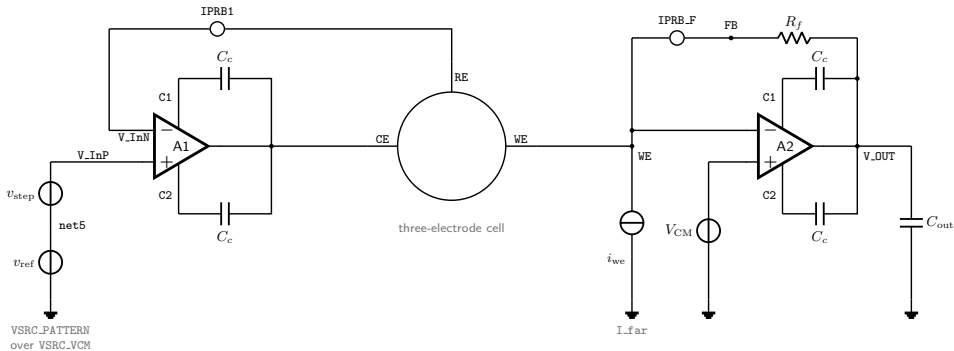
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A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we} R_f$, both polarities.

Analog top level

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Virtuoso export of the analog top level: four channels, shared bias, test bus

The electrode model, in Verilog-A

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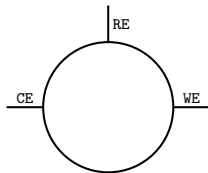
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R_u , R_{ce} , C_{dl} lumped; the faradaic branch Z_f is the Verilog-A element.

i_f faradaic current
 η overpotential, $E - E^{0'}$
 f F/RT
 $C_{O,R}(0, t)$ surface conc. of O, R
 C^* bulk conc.
 δ Nernst layer

Kinetics $i_f = nFAk^0 [C_R(0, t) e^{(1-\alpha)nf\eta} - C_O(0, t) e^{-\alpha nf\eta}]$
 Transport $\partial C / \partial t = D \partial^2 C / \partial x^2$, finite-volume, 28 expanding slabs per species
 Boundaries flux = $\pm i_f / nFA$ at the surface; $C(\delta, t) = C^*$ at the Nernst layer

Parameter	Symbol	Default	Unit	Quantity
E0	$E^{0'}$	0.0	V	formal potential of the couple
nel	n	1	—	electrons transferred
alpha	α	0.5	—	transfer coefficient
k0	k^0	0.1	cm s^{-1}	standard heterogeneous rate constant
Dif	D	5×10^{-6}	$\text{cm}^2 \text{s}^{-1}$	diffusion coefficient, both species
Area	A	0.01	cm^2	electrode area
Cox	C_O^*	1×10^{-6}	mol cm^{-3}	bulk concentration, oxidised species
Cred	C_R^*	0	mol cm^{-3}	bulk concentration, reduced species
Temp	T	298.15	K	interface temperature
expmax	—	40	—	exponent clamp on the kinetic terms
gpar	—	1×10^{-15}	S	DC shunt across the terminals

Equations: Bard–Faulkner [13]. Discretisation: Britz–Strutwolf, Feldberg [17, 18]. Precedent for Verilog-A electrode co-simulation: Jog et al. [19].

Emulating the four markers

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Marker	Reporter	$E^{0'}$ (V)	Peak (nA)	C_R^* (μ M)	Cut-off
CA-125	methylene blue	-0.15	856	279	35 U/mL [5, 8]
HE4	ferrocene	+0.10	1311	394	70 pmol/L [6]
CA19-9	TMB / HRP [20]	+0.25	1659	458	37 U/mL [5]
CEA	anthraquinone	+0.40	939	303	5 ng/mL [5, 14]

$E^{0'}$ reporter formal potential, i.e. peak position; chosen in the -0.6 to $+0.4$ V window [22, 23], 150–250 mV apart, not measured per reporter

Peak inside 50 nA to 5 μ A on screen-printed electrodes [14, 15, 21]; C_R^* fitted to reach it

k^0 0.1 cm/s, reversible; nothing published on carbon for these markers

Shared $\alpha = 0.5$, $n = 1$, $D = 5 \times 10^{-6}$ cm²/s, $A = 0.01$ cm², 37 °C, $R_f = 560$ k Ω

Representative values, not marker-measured: these slides demonstrate the channel, not an assay.

Channel range, assay range, and where the panel was placed

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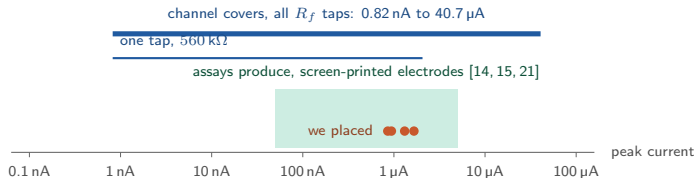
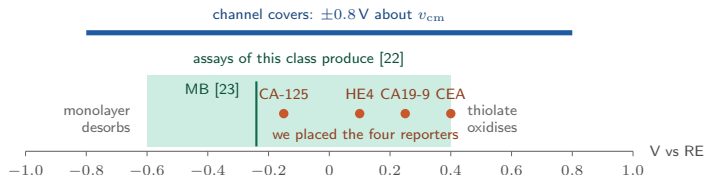
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Potentials: spaced 150–250 mV for one-sweep resolution inside the reporter window, not per-reporter values. Peak heights: inside the published envelope and one tap's full scale, then C_R^* fitted. Silicon is unaffected: the channel covers the assay band with 0.2–0.4 V and a decade of current to spare, so moving the dots changes what the simulation shows, not what the chip can measure.

Exercising the cell: differential-pulse voltammetry

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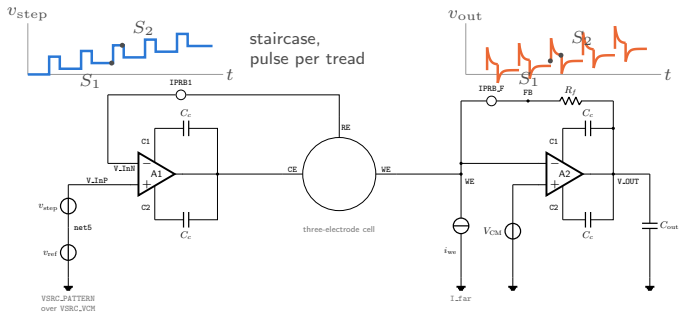
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Excite

DAC: staircase with a pulse on every tread, applied at RE by A1

Sense

$$v_{out} = v_{cm} - i_{WE} R_f$$

Sample

SAR at S_1 (before the pulse) and S_2 (end of the pulse), once charging has decayed

Resolve

firmware: $\Delta I = I(S_2) - I(S_1)$ vs E ; peak position: marker, peak height: concentration

Four markers resolved in one differential-pulse sweep

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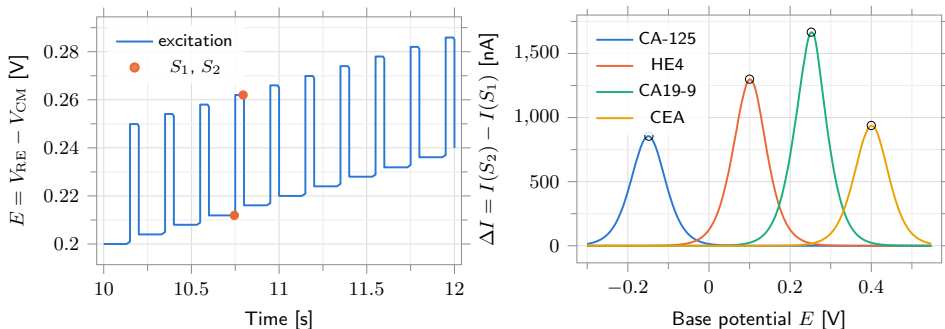
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*schematic sim

Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

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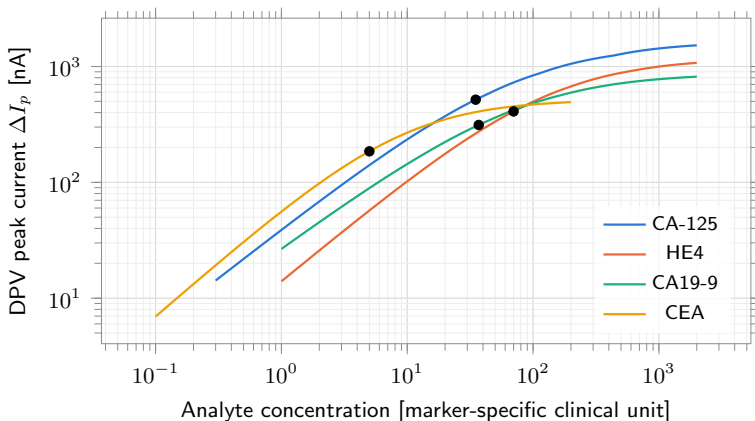
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*schematic sim

CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Cyclic voltammetry through the assembled channel

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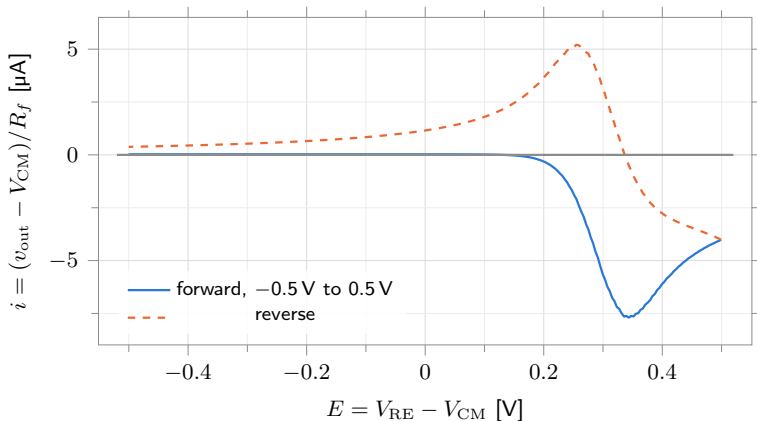
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*schematic sim

Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

How impedance is measured: the same loop, no extra hardware

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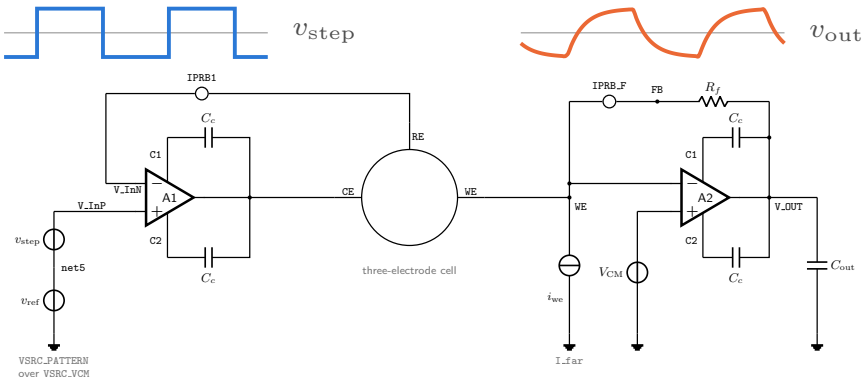
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- | | |
|----------|--|
| Excite | DAC: a small two-code square about the DC setpoint v_{ref} |
| Sense | v_P , the commanded potential (RE tracks it through A1; RE tap pending); $i_{WE} = -v_{out}/R_f$ |
| Digitise | one SAR, both records interleaved; firmware I/Q MAC over M periods |
| Solve | $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$, or ratio to Z_{cal} (R_f , LSB, $4/\pi$ cancel) |

Magnitude and phase from codes: I/Q demodulation

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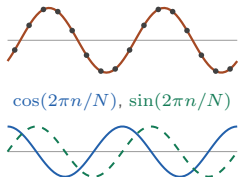
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References

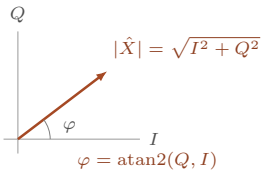
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record c_n , N per period, M periods



$$I = \frac{2}{MN} \sum c_n \cos \frac{2\pi n}{N}$$

$$Q = \frac{2}{MN} \sum c_n \sin \frac{2\pi n}{N}$$



1. Records two, interleaved on the one SAR: v_P (the DAC command, slot 2; RE tracks it through A1) and v_{out}
2. Phasor per record, $\hat{X} = I - jQ$; the sine reference selects one frequency, so harmonics, DC and noise average out
3. Skew v_{out} is sampled half a conversion late: rotate $\hat{V}_{OUT}$ by $180^\circ f/f_s$
4. Divide $|Z| = R_f |\hat{V}_P|/|\hat{V}_{OUT}|$, $\angle Z = \varphi_P - \varphi_{OUT} + 180^\circ$; LSB and $4/\pi$ cancel
5. Ratio same on Z_{cal} at the same f and gain code: R_f , its pole and the DAC droop cancel

The demodulation, in full (1/2): from codes to a phasor

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Excite	$v_{\text{ref}}(t) = V_{\text{CM}} + \frac{A}{2} \text{sq}(2\pi ft)$, two DAC codes; fundamental $\frac{4}{\pi} \frac{A}{2} \cos 2\pi ft$, odd harmonics at $3f, 5f, \dots$
Sample	$f_s = Nf$, N conversions per period, M whole periods: $c_n = \lfloor v(nT_s)/\text{LSB} \rfloor$, $n = 0 \dots MN - 1$, one such record per channel
Model	$c_n = C_0 + A_1 \cos\left(\frac{2\pi n}{N} + \varphi\right) + \sum_{k \geq 2} A_k \cos\left(\frac{2\pi kn}{N} + \varphi_k\right) + w_n$
Correlate	$I = \frac{2}{MN} \sum_{n=0}^{MN-1} c_n \cos \frac{2\pi n}{N}$, $Q = \frac{2}{MN} \sum_{n=0}^{MN-1} c_n \sin \frac{2\pi n}{N}$
Orthogonality	over whole periods $\sum \cos^2 = \frac{MN}{2}$, $\sum \cos \cdot \sin = 0$, $\sum \cos \frac{2\pi n}{N} \cos \frac{2\pi kn}{N} = 0$ for $k \neq 1$, $\sum \cos = 0$
Result	$I = A_1 \cos \varphi$, $Q = -A_1 \sin \varphi$, $\hat{X} = I - jQ = A_1 e^{j\varphi}$; DC and every harmonic vanish $ \hat{X} = \sqrt{I^2 + Q^2}$, $\varphi = \text{atan2}(-Q, I)$
Noise	white w_n , variance σ_w^2 : $\sigma_{ X }^2 = 2\sigma_w^2/MN$, i.e. 6.02 dB per $4 \times$ periods; $\sigma_{ Z }/ Z $ from 1.20 % at $M=1$ to 0.009 % at $M=16384$ ($N=0$ tap)

The demodulation, in full (2/2): from two phasors to R_{ct} and C_{dl}

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Records	$\hat{V}_P$ from the potential channel (the DAC command; RE tracks it through A1), $\hat{V}_{OUT}$ from the TIA output, interleaved on one SAR
Skew	v_{out} samples sit at $(n + \frac{1}{2})T_s$, a lag of $\pi f/f_s$: $\hat{V}_{OUT} \leftarrow \hat{V}_{OUT} e^{+j\pi f/f_s}$
Impedance	$\hat{Z} = \frac{\hat{V}_P}{\hat{I}_{WE}} = -R_f \frac{\hat{V}_P}{\hat{V}_{OUT}}, \quad Z = R_f \frac{ \hat{V}_P }{ \hat{V}_{OUT} }, \quad \angle Z = \varphi_P - \varphi_{OUT} + \pi$ the LSB and the square's $\frac{4}{\pi} \frac{A}{2}$ are in both records and cancel in the ratio
Calibrate	$\hat{Z} = Z_{cal} \frac{\hat{V}_P \hat{V}_{OUT}^{cal}}{\hat{V}_{OUT} \hat{V}_P^{cal}}$ at the same f and gain code: R_f , its pole and the DAC carry droop cancel
Extract	$R_u = \Re \hat{Z}$ at the top of the band; $Y = \frac{1}{\hat{Z} - R_u} = \frac{1}{R_{ct}} + j\omega C_{dl}$; $R_{ct} = 1/\Re Y, \quad C_{dl} = \Im Y/\omega$
Cost	$2MN$ multiply-accumulates per frequency point, on the channel hart or the NPU

Impedance spectroscopy on the same channel

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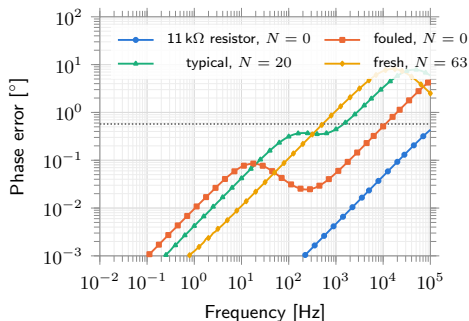
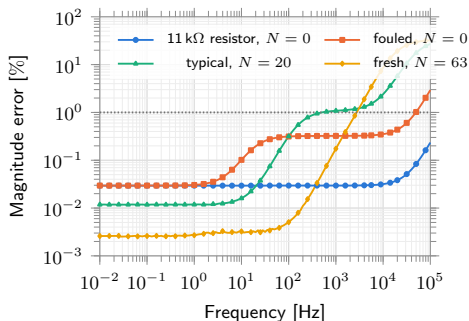
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Excitation	Demodulation	Accuracy	$ Z $ range
2-code DAC square	firmware, channel hart	$\pm 1\%$, $\pm 0.6^\circ$ to 5.05 kHz; $\pm 5\%$ to 26.5 kHz	246 Ω – 12.1 M Ω

*schematic sim

No EIS engine. Impedance is a ratio of two readings through one converter; the DAC drops out of the accuracy budget.

The class-AB amplifier: one cell, both roles

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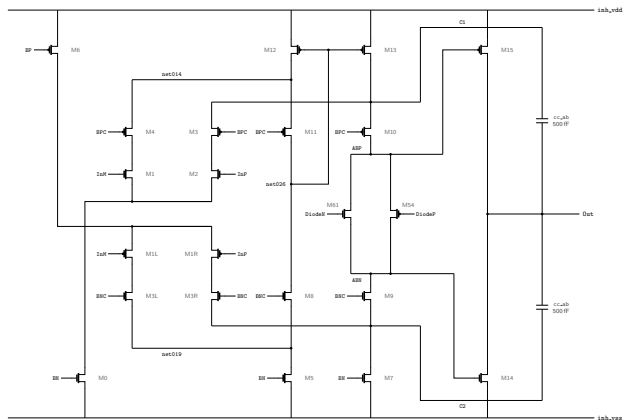
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complementary
input pairs and
their cascodes

diode-connected
summing branch

class-AB
drive

output

testbench
compensation

Bipolar accuracy through zero

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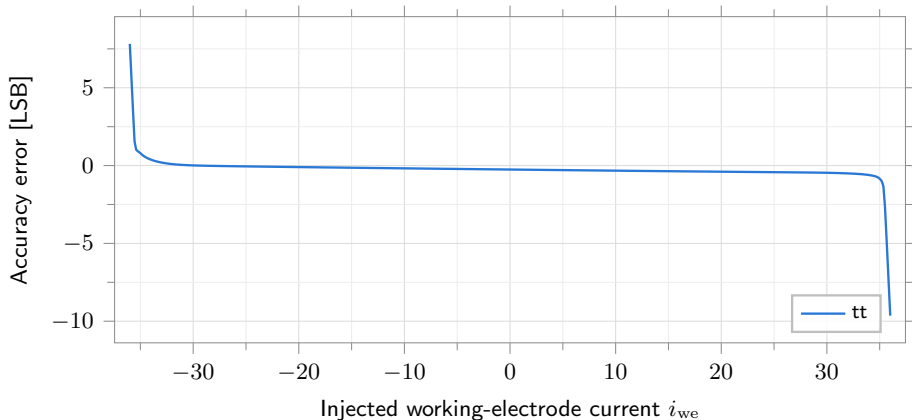
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*schematic sim

No kink at the crossing: class-AB output, no dead zone. This is the measurement rev-1's PMOS mirror could not make.

Programmable feedback resistor: one string, 320 codes

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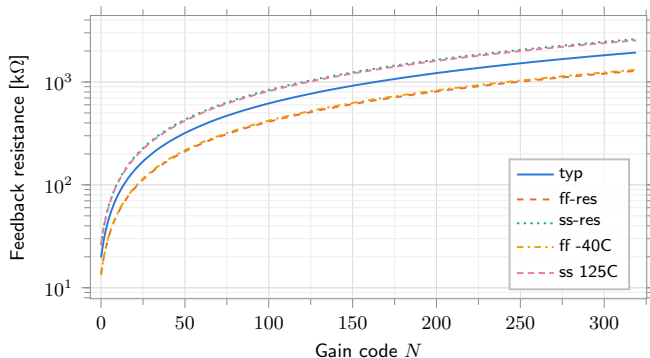
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$$R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$$

N	R_f	Full scale
0	19.6 kΩ	±40.7 μA
63	397 kΩ	±2.02 μA
319	1.93 MΩ	±0.41 μA
LSB at 319		0.824 nA

Strictly monotonic at every corner, including the four thermometer seams. Maximum gain is noise-limited, not quantisation-limited.

10-bit SAR ADC, extracted netlist

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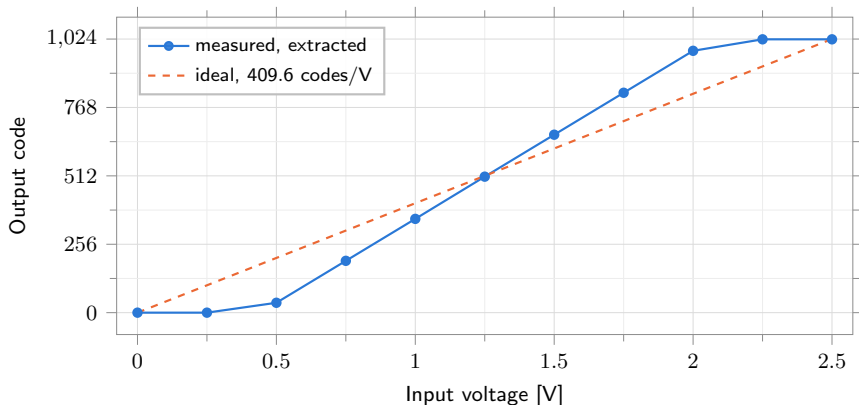
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■ extracted

netlist

23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

Block results: corners, with mismatch σ

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Parameter	Unit	tt	ff	fs	sf	ss	Spread	σ (MC)
Class-AB amplifier (A1 and A2)								
Open-loop gain	dB	82.77	79.88	83.14	83.37	84.94	5.06	1.06
Phase margin	deg	73.52	74.70	74.04	72.86	73.34	1.84	1.42
Dominant pole	kHz	1.592	2.286	1.538	1.494	1.225	1.061	0.276
Gain-bandwidth product	MHz	22.78	23.19	22.89	22.94	22.59	0.60	3.57
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	0.193	5.201
Programmable feedback resistor^a								
R at $N=0$	k Ω	19.64	—	—	—	—	—	0.071
R at $N=319$	M Ω	1.931	—	—	—	—	—	0.0007
Seam step 63 $\rightarrow$ 64	k Ω	7.42	—	—	—	—	—	0.504
10-bit SAR ADC, extracted^b								
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13	0.60	—
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920	0.0015	—
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399	0.0009	—
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286	0.0015	—
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484	0.018	—
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302	0.025	—
Analog supply current	μ A	6.36	4.79	5.45	8.44	15.43	10.64	—
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5	31.4	—
12-bit R-2R reference DAC, ideal rail								
Full-scale output	V	2.499388	2.499388	2.499388	2.499388	2.499388	0	0.000001
LSB size	μ V	610.351	610.351	610.351	610.351	610.351	0	0.0003
INL, most positive	LSB	0.113	0.101	0.100	0.127	0.130	0.030	0.610
INL, most negative	LSB	-0.376	-0.338	-0.390	-0.362	-0.414	0.076	0.620
DNL, most positive	LSB	0.002	0.001	0.002	0.002	0.002	0.000	1.128
DNL, most negative	LSB	-0.423	-0.374	-0.424	-0.424	-0.479	0.105	1.415

σ : 200 samples, process + mismatch, nominal corner. Corner runs hold the passive models at typical, so resistor and ladder spreads are lower bounds. ^aThe TRM carries R_f corners as a figure (the R_f slide), not a table. ^bExtracted netlist; the extraction carries no mismatch statistics, so no σ . Amplifier and DAC: schematic sim*; SAR: extracted.

One AFE channel, as simulated

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Parameter	Value	Condition
CE compliance	13.2 mV – 2.480 V	2.5 V rail, corners
Cell current delivered	$\pm 52.5 \mu\text{A}$	12 k Ω cell
TIA compliance	$\pm 0.80 \text{ V}$ about v_{cm}	
Feedback resistor	19.6 k Ω – 1.93 M Ω , 320 codes	monotonic at every corner
Full-scale current	$\pm 40.7 \mu\text{A}$ ($N=0$) to $\pm 0.41 \mu\text{A}$ ($N=319$)	SAR window $\cap$ TIA compliance
Current quantum	0.824 nA at maximum gain	below the 2 nA noise budget
Bipolar accuracy	0.25–0.38 LSB worst	560 k Ω , 1 M Ω electrode, corners
Control-loop PM	79–80°	five corners, into a cell
TIA-loop PM	89–93°	12-cell electrode grid
Input current noise	341 pA standalone, 3.6–21 nA into an electrode	0.1–100 Hz, 560 k Ω
Zero-current offset	$\sigma(v_{\text{re}}) = 5.72 \text{ mV}$, gain-independent	200 MC samples, calibratable
Quiescent current	197–250 μA per channel	amplifiers + local bias

* schematic sim ■ Monte Carlo, 200 samples

Pre-layout, electrode held nominal: no row includes electrode-to-electrode spread.

Castalia at a glance

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Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	10.72 mW post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Digital, as closed

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Whole chip, post-route

Internal	3.78 mW
Switching	2.33 mW
Leakage	4.61 mW

Total	10.72 mW
--------------	-----------------

One channel tile

Total	1.761 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726
Aggregate	1 314 802	1.260

Timing at 25 MHz

reg2reg WNS	+99 ps
Leakage share	43 %

■ hardened flow, post-route

SS 0.9 V 125 °C, statistical activity 0.2. Gating one tile removes 1.76 mW; the always-on band pays nothing.

Power gating: one switched rail per channel tile

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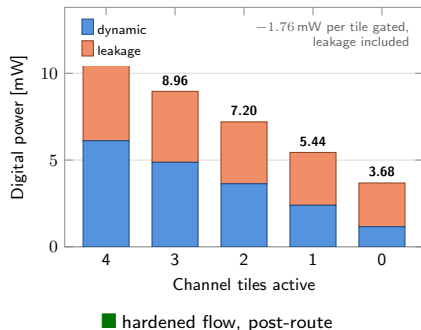
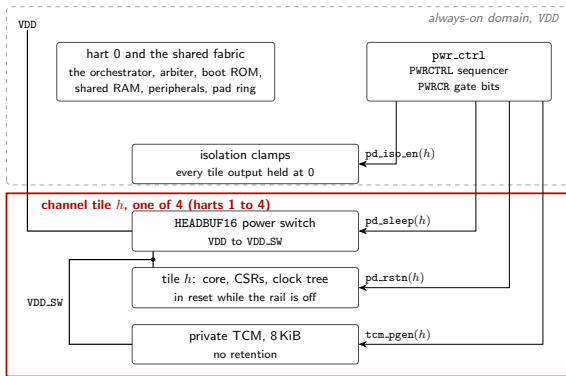
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Cold gating: the header cuts the rail, so a parked tile draws neither dynamic nor leakage power. Each tile gated removes 1.76 mW; the always-on band keeps 3.68 mW.

Against the field: the ISCAS'26 table, plus Castalia

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Parameter	Castalia (this work)	Myshkin [9]	Chen '19 [10]	Ghoreishizadeh '17 [11]	Lin '23 [12]
Modality	CA/CV/DPV/EIS	CA/CV	CA/CV	CA/CV	CV/EIS
Channels	4, simultaneous	1	1	1, differential	1
Current direction	bipolar	unipolar	bipolar	bipolar	bipolar
Sensitivity (pA)	341 ^a	56.5 ^b	41	0.47	1.7
$I_{\max}$ (μA)	±40.7	+20	±5	±20	-48/+16
Power (mW)	3.68 ^c	3.0 ^d	0.016	9.3	1.0
Process (nm)	65	65	180	350	180
On-chip processor	5 RISC-V harts	1 RISC-V hart	none	none	none
On-chip ML	NPU, 4 modes	NPU, 1 mode	none	none	none

Castalia column is best case and pre-silicon. ^aAFE only: input-referred rms, 0.1–100 Hz, maximum gain, ideal R_f , no electrode; the converter LSB is 0.82 nA. ^bAFE only; ADC quantisation limits the system to 153 pA. ^cDigital only: hart 0 awake, all four channel tiles gated, post-route SS 125 °C 25 MHz; one channel measuring adds 1.76 mW (tile) + 1.14 mW (AFE). ^dMeasured: 2.8 mW digital at 24 MHz + 258 μW AFE.

Still the only entries with a processor and an ML engine on the die; Castalia adds concurrency, both current polarities and EIS.

The fusion runs on the NPU

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$$s = \sigma \left(\sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

Input	peak current, peak potential, baseline $\times 4$
Mode	0, dense layer, one sigmoid
Output	one risk score
Weights	trained off-chip, staged by firmware

Per site

Hart	1, ownership enforced in hardware
Potential	own 12-bit DAC
Converter	own 10-bit SAR
Power	own MTCMOS domain

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

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Backup: electrode model validation

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Scan rate mV s^{-1}	i_p Randles–Ševčík μA	i_p model μA	Error %	Error, $C_{\text{dl}} \rightarrow 0$ %	ΔE_p mV
25	0.951	0.952	+0.08	−0.45	58.78
50	1.345	1.349	+0.32	−0.42	59.85
100	1.902	1.915	+0.67	−0.38	60.41
200	2.690	2.721	+1.16	−0.33	62.33
400	3.804	3.875	+1.86	−0.25	64.74

Peak current against Randles–Ševčík and ΔE_p against Nicholson [13], reversible couple, $C_{\text{dl}} = 0$ column isolates the faradaic model.

Backup: USPSTF screening grades

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Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021) prostate PSA 55–69 (2018)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.